

Combinational Circuits

- Adders
- Subtractors
- Comparators
- Encoders
- Decoders
- Multiplexers
- Demultiplexers

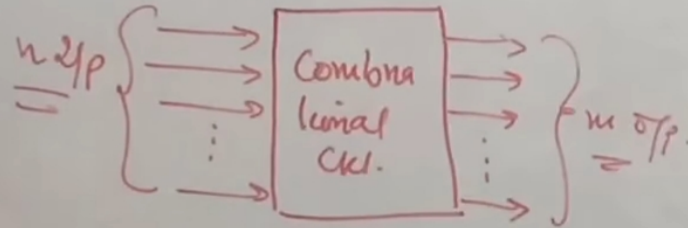
Digital Circuits

Combinational
Circuits

Sequential
Circuits

{ - O/P depends only
on present I/P

- O/P depends upon present I/P
as well as previous O/P



Combinational Circuits

Digital Circuits

Combinational Circuits

- O/P depends only on present.

- Adder, Sub, Comparators, Multiplexers, Demux, PLA, PAL, Rom.

Adder = $\left\{ \begin{array}{c} 1 \\ + 0 \end{array} \right\}$ P.A.
1 → o/p

Sequential Circuits

- O/P depends upon present o/p as well as previous o/p

Adders
Factories
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ultiplexers
ultiplexers

Combinational Circuits

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Digital Circuits

Sequential Circuits

- O/P depends upon present I/P as well as previous O/P
→ Flip-flops, Counters, Registers

Counter →
 $4 + 1 = 5$
 $5 + 1 = 6$
 $6 + 1 = 7$

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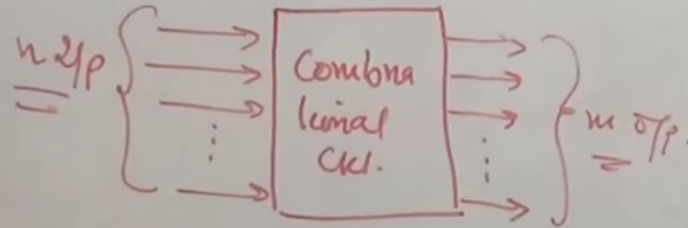
Digital Circuits

Combinational Circuits

Sequential Circuits

{ - O/P depends only on present I/P

- O/P depends upon present I/P as well as previous O/P



- In a Sequential Circuit, the "previous output" is stored in a physical memory element (like a Flip-Flop) and fed back into the circuit after a clock pulse.
- The circuit literally "remembers" what happened in the past.
- In combinational ckt., there's no such feature.

